

Titan Protocell Catalysis Framework

Latex-form rate model for solvent, surface, and catalyst effects

Titan surface conditions: $T \simeq 94\text{--}100\text{ K}$, $P \simeq 1.5\text{ bar}$

1. General effective rate expression

For a reaction r occurring in environment e , the effective rate constant is written as

$$k_r^{(e)} = k_{r,0} \exp \left[-\frac{E_{a,r}}{R} \left(\frac{1}{T_e} - \frac{1}{T_{\text{ref}}} \right) \right] \Phi_r^{(e)} \Gamma_r^{(e)} \Lambda_r^{(e)} \quad (1)$$

Here:

- $k_{r,0}$ is the reference rate constant at T_{ref} .
- $E_{a,r}$ is the activation energy.
- $\Phi_r^{(e)}$ is the solvent/activity factor.
- $\Gamma_r^{(e)}$ is the surface-localization factor.
- $\Lambda_r^{(e)}$ is the catalytic enhancement factor.

The Arrhenius term controls the strong low-temperature suppression at Titan conditions. The other factors should not be treated as arbitrary fitting constants. They represent physical availability, localization, and catalysis.

2. Solvent/activity factor

For a reaction



the solvent factor can be defined as

$$\Phi_r^{(e)} = \prod_{i \in \text{reactants}} \left(a_i^{(e)} \right)^{\nu_i} \quad (3)$$

where $a_i^{(e)}$ is the activity of reactant X_i in environment e .

A practical simulation-level approximation is

$$a_i^{(e)} = S_i^{(e)} x_i^{(e)} \quad (4)$$

so that

$$\Phi_r^{(e)} = \prod_i \left(S_i^{(e)} x_i^{(e)} \right)^{\nu_i} \quad (5)$$

where:

- $S_i^{(e)}$ is a dimensionless solubility or partition coefficient.
- $x_i^{(e)}$ is the concentration or mole fraction of species i in environment e .
- ν_i is the stoichiometric coefficient of reactant i .

Interpretation:

- Methane-rich liquid favors small nonpolar hydrocarbons such as CH_4 , C_2H_6 , C_2H_2 .
- Ethane-rich liquid is generally better for larger hydrocarbons and less volatile organics.
- Nitrile-rich and polymeric species may have low solubility in bulk hydrocarbon liquid and therefore accumulate on surfaces or evaporites.

3. Surface-localization factor

For a reaction requiring two adsorbed reactants A and B , define

$$\Gamma_r^{(e)} = \left(\frac{A_{\text{surf}}}{V} \right) \theta_A \theta_B \quad (6)$$

where:

- A_{surf}/V is the surface area per volume.
- θ_A and θ_B are fractional surface coverages.

For one adsorbed reactant A reacting with one dissolved species B , use

$$\Gamma_r^{(e)} = \left(\frac{A_{\text{surf}}}{V} \right) \theta_A C_B \quad (7)$$

where C_B is the concentration of B near the surface.

Surface coverages can be modeled with a Langmuir adsorption expression:

$$\theta_i = \frac{K_i C_i}{1 + \sum_j K_j C_j} \quad (8)$$

where K_i is the adsorption coefficient of species i and C_i is its concentration.

The factor $\Gamma_r^{(e)}$ is large when the system has high surface area, strong adsorption, and high surface coverage. This is essential for Titan because many low-temperature reactions are unlikely in dilute bulk liquid but may proceed on tholin, ice, mineral, evaporite, or polymer surfaces.

4. Catalytic enhancement factor

The catalyst factor describes enhancement by active sites. A simple expression is

$$\Lambda_r^{(e)} = 1 + \sum_c \eta_{rc} \Theta_c \quad (9)$$

where:

- c labels catalysts such as tholins, mineral sites, ice surfaces, HCN polymers, or PAH-like fragments.
- η_{rc} is the catalytic strength of catalyst c for reaction r .
- Θ_c is the active-site availability of catalyst c .

The active-site availability can be written as

$$\Theta_c = \frac{N_{c,\text{active}}}{N_{c,\text{total}}} \quad (10)$$

or, for a concentration-based model,

$$\Theta_c = \frac{K_c C_c}{1 + K_c C_c} \quad (11)$$

This lets different catalysts affect different reaction classes. For example:

- tholin surfaces: radical recombination and nitrile polymerization;
- mineral or ice surfaces: adsorption, concentration, and orientation of reactants;
- HCN polymers: autocatalytic HCN oligomer growth;
- PAH/tholin fragments: acetylene addition and hydrocarbon chain growth.

5. Recommended implementation form

For code, keep the factors modular:

$$k_r^{(e)} = k_{r,0} F_T \Phi \Gamma \Lambda, \quad (12)$$

$$F_T = \exp \left[-\frac{E_{a,r}}{R} \left(\frac{1}{T_e} - \frac{1}{T_{\text{ref}}} \right) \right], \quad (13)$$

$$\Phi = \prod_i \left(S_i^{(e)} x_i^{(e)} \right)^{\nu_i}, \quad (14)$$

$$\Gamma = \left(\frac{A_{\text{surf}}}{V} \right) \prod_i \theta_i, \quad (15)$$

$$\Lambda = 1 + \sum_c \eta_{rc} \Theta_c. \quad (16)$$

In pseudocode:

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k_eff = k0
k_eff *= arrhenius_factor(T, Ea, T_ref)
k_eff *= solvent_activity_factor(reaction, solvent)
k_eff *= surface_localization_factor(reaction, surface)
k_eff *= catalytic_enhancement_factor(reaction, catalysts)

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6. Physical interpretation for Titan protocells

A polymer-rich Titan protocell analog should not be modeled as a lipid vesicle. It is better represented as an aggregate or compartment whose formation is controlled by polymer accumulation, low solubility, adsorption, and surface catalysis.

The key condition for compartment formation is

$$\boxed{M_{\text{polymer}} > M_{\text{agg}}^*} \quad (17)$$

where M_{agg}^* is an aggregation threshold depending on solvent composition, temperature, and polymer type.

A possible growth equation is

$$\frac{dM_{\text{polymer}}}{dt} = \sum_{r \in \text{polymerization}} y_r R_r^{(e)} - k_{\text{frag}} M_{\text{polymer}} - k_{\text{loss}} M_{\text{polymer}}, \quad (18)$$

where y_r is the polymer yield of reaction r , $R_r^{(e)}$ is the reaction rate in the local environment, k_{frag} is a fragmentation rate, and k_{loss} is a degradation or dissolution rate.

7. Selection variables

For evolutionary dynamics, define a compartment fitness as

$$\boxed{\mathcal{F} = w_1 \tau_{\text{survival}} + w_2 Y_{\text{polymer}} + w_3 R_{\text{solute}} + w_4 S_{\text{structural}} - w_5 D_{\text{photo}}} \quad (19)$$

where:

- τ_{survival} is survival time;
- Y_{polymer} is polymer growth yield;
- R_{solute} is retention of useful solutes such as HCN, C_2H_2 , HC_3N , and CN-bearing fragments;
- $S_{\text{structural}}$ is aggregate stability;
- D_{photo} is photochemical or radical damage.

This gives a concrete way to model selection without claiming Earth-like biology. Evolution acts on persistence, catalytic efficiency, solute retention, and structural stability.

8. Summary

The factors in Eq. (1) have the following roles:

Factor	Meaning	Main variables
F_T	thermal activation	T, E_a
Φ	solvent availability	solubility, partitioning, activity
Γ	surface localization	surface area, adsorption, coverage
Λ	catalysis	active sites, selectivity, catalytic strength

This rate structure is suitable for Titan because it separates the three critical non-Arrhenius effects: solubility in methane/ethane liquids, concentration on surfaces, and catalytic enhancement by tholins, evaporites, minerals, ice, or polymeric aggregates.